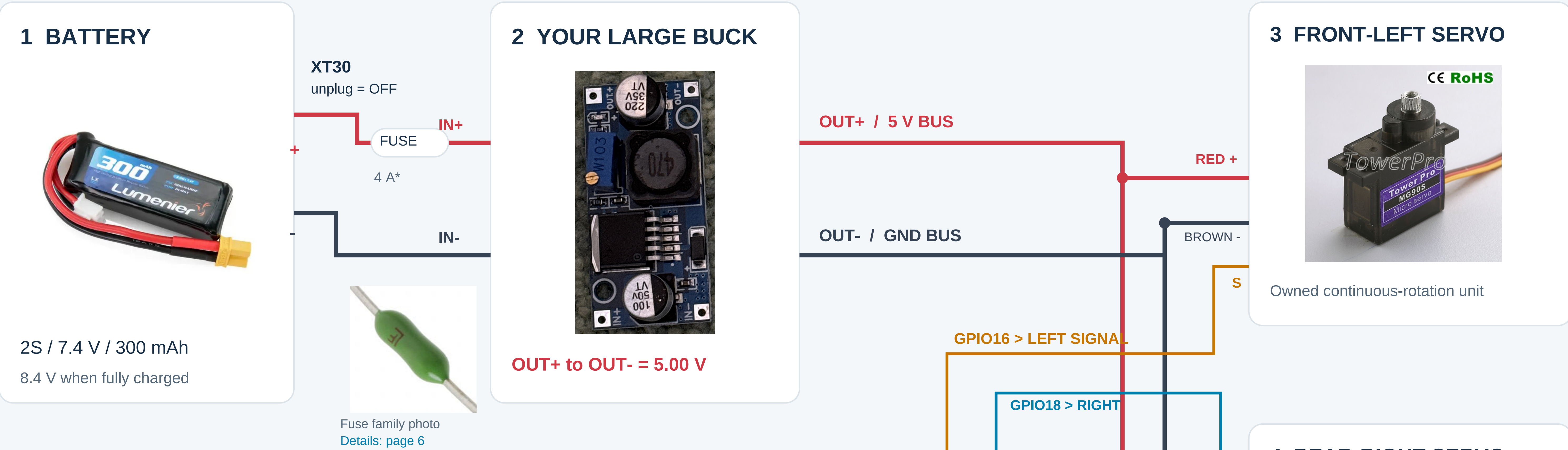
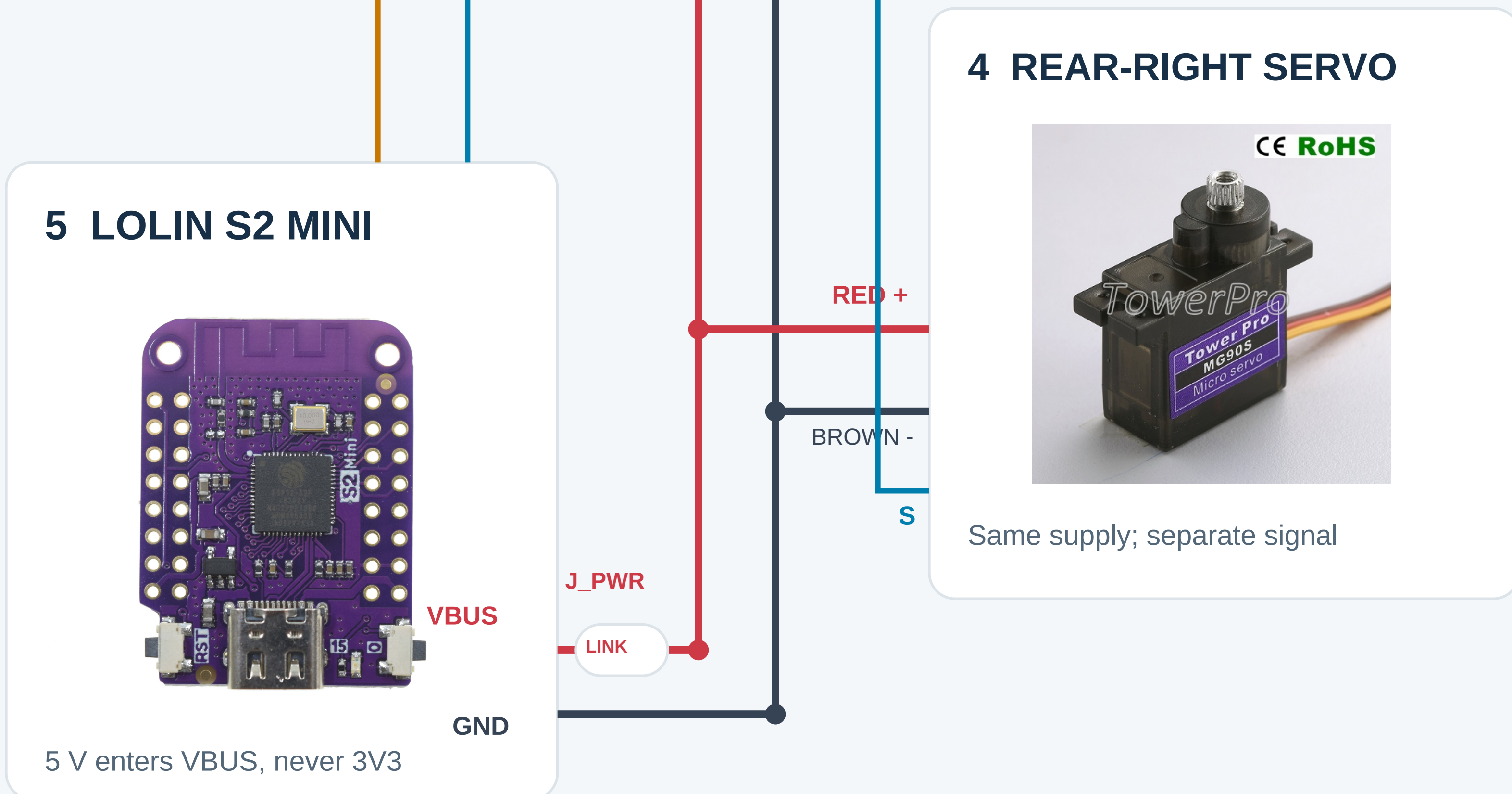
Connect the car

Read the named terminals here; page 2 locates the actual solder pads. All grounds connect together. Crossings connect only where marked with a dot.





At the 5 V / GND split

At the split: **220 uF / 10 V electrolytic** (+ to 5 V; striped negative to GND). At the S2, **after J_PWR:** add **100 nF ceramic across VBUS and GND** with short leads. Preserve the buck's onboard capacitor.

Manual battery checks: page 4

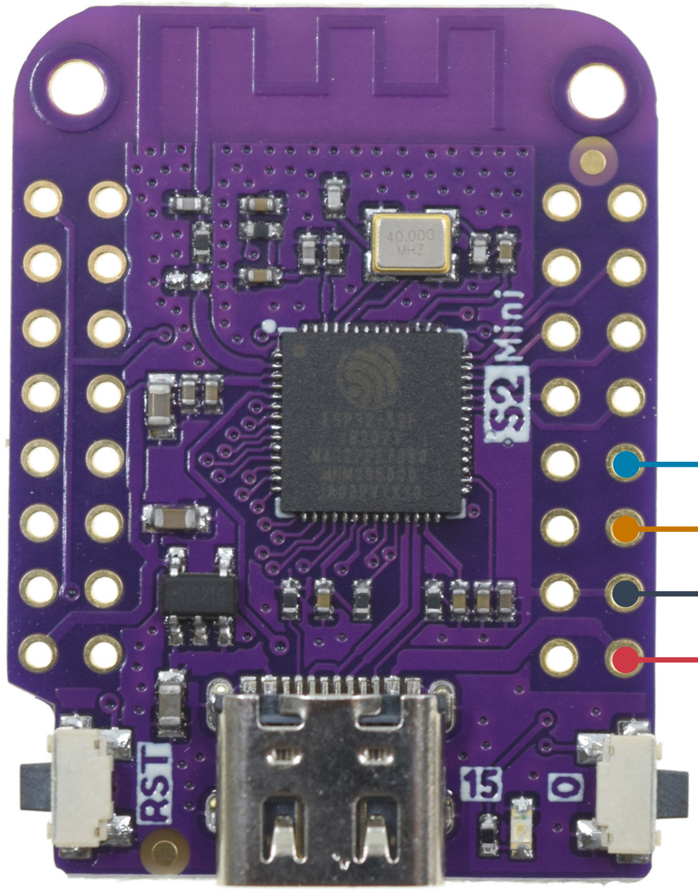
No sensing perfboard or GPIO3/GPIO7 wires. The car has **no battery reading, alarm or low-voltage stop**. Check each cell with a meter between short runs and unplug when parked.

* Fuse rating is provisional. Photos show reference servos; wire functions, not plug orientation, define the connections.

Exactly where each wire lands

Component side facing you. S2 antenna at the top; USB-C at the bottom. Photos are not mirrored.

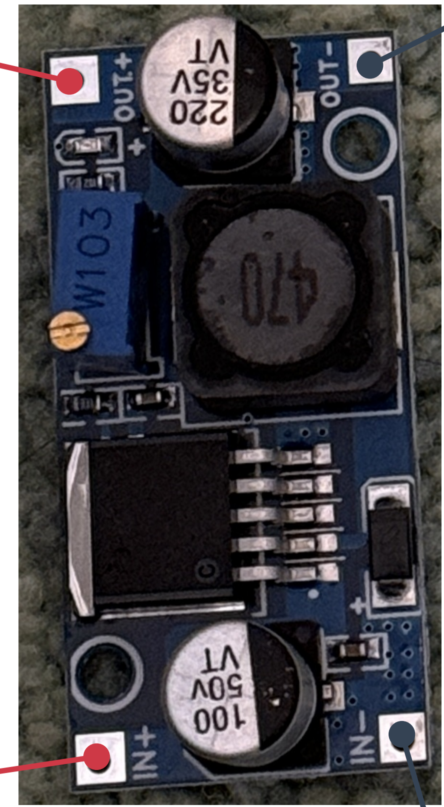
LOLIN S2 MINI v1.0.0 REFERENCE



Only four S2 connections: VBUS, GND, GPIO16 and GPIO18.

Use GPIO labels. Inner-row GPIO17 is NOT the right servo pin.

YOUR LARGE BUCK: SAME ORIENTATION



Blue trimmer sets output. Measure OUT+ to OUT- and set **5.00 V with all loads disconnected**. IC marking is not readable: confirm the board type.

Soldered wiring + removable connectors

Bare S2 holes and converter pads need soldered wires or soldered headers. Use 22 AWG or thicker for the battery and main 5 V / GND paths. Servo red normally means supply; brown/black ground; orange/yellow signal. Verify your servo wires before fitting a 3-pin extension. Split its three wires to the correct nets - do not plug the whole servo lead onto random adjacent S2 pins.

Battery, plug and converter choice

Reuse the larger buck for this prototype. A custom power PCB is not required.

BUY: LUMENIER 300

2S 75C LiPo, XT30
48 x 17 x 12 mm
18 g; 7.4 V nominal
8.4 V fully charged

$13.49; listed available
at the research snapshot.

RaceDayQuads product / purchase page

YOUR YELLOW CONNECTORS

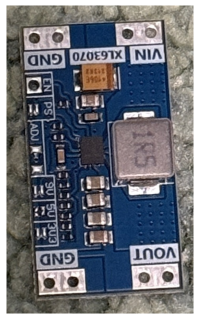
The photo establishes the XT family, but does not distinguish **XT30 from XT60**. Read the molded marking.

XT30: use the mating harness half.
XT60: buy an XT30 mating pigtail; a bulky adapter also works but increases footprint.

Check + / - with a meter. Housing shape or wire color alone is not proof of polarity.

Why the large buck wins here

A 2S pack feeds a step-down converter naturally. Allowing 1.2 A per servo plus 0.4 A for the S2 gives a **provisional 2.8 A at 5 V**. Actual servo peaks are unknown. The LM2596 IC is rated 3 A with suitable parts and cooling; this particular board must still pass load and temperature checks at 8.4 V and 7.0 V.



Why not the small XL63070 board?

It appears to be a TPS63070-style buck-boost module; the IC is unconfirmed. TI specifies 2 A output at 4 V in / 5 V out. That is below our provisional load. A 1S design at 14 W could also need about 5 A near 3.3 V. Smaller hardware needs current measurements first.

Battery handling and fit

12 mm is the published nominal pack height, not measured installed height: leave room for padding and lead bends. This hobby pack has no verified protection PCB. Neither owned converter provides charging, balancing or a suitable LiPo cutoff. Use a 2S balance charger and unplug the pack after each run.

Manual battery checks

The perfboard is removed. GPIO3 and GPIO7 stay unwired. The car cannot detect a low battery.

No automatic low-voltage protection

The fuse opens on sufficiently large overcurrent, **not** low voltage. Firmware has no pack reading, low-battery alarm or voltage-triggered motion stop. A joystick stop leaves the electronics powered. **Unplug XT30 whenever parked.**

Measure the two cells separately

Unplug XT30 from the car. Set the multimeter to **DC volts**, leads in **COM and V**, never the current socket/range. Prefer an insulated mating balance breakout.

The three balance taps are pack negative, cell midpoint and pack positive. Each adjacent pair measures one cell; the two end taps measure the whole pack.

Verify tap order and polarity on your pack. Do not assume left/right orientation or colors. Never bridge adjacent contacts with metal probe tips.

Short, attended trials

Start with a correctly balance-charged pack. Check both cells before and between very short runs while establishing runtime.

End the session at or below 3.7 V per cell at rest, or earlier if the pack instructions require. This is an early project stopping point, not a battery minimum or a guarantee against dips during use. At 3.5 V or lower, do not do another run.

Unplug promptly for abnormal heat, puffing, slowing or resets. Do not use those symptoms as the normal battery gauge.

What a multimeter cannot do

Periodic checks cannot catch a weak cell or rapid voltage drop between measurements. Total pack voltage can hide an individually depleted cell. No safe fixed runtime has been measured for this car.

Before Wi-Fi updates: use a freshly balance-charged pack and lift the wheels. Firmware does not check voltage before or during upload. Longer unattended operation needs a different protection plan.

Build, check and use

No custom PCB. This is a soldered prototype harness, not a tested plug-and-play electrical assembly.

Wire and check in this order

1. Battery unplugged: solder the mating XT30 lead through a provisional 4 A inline fuse to IN+; negative to IN-. Insulate every joint.

2. Leave all loads disconnected. Power the buck, read OUT+ to OUT- with a multimeter and adjust to 5.00 V. Unplug before adding wires.

3. Split OUT+ and OUT- into separate servo and S2 branches. Add the capacitors. Fit J_PWR as a removable insulated connector in the S2 VBUS feed.

4. Leave GPIO3 and GPIO7 unwired. There is no sensing circuit. Check both cells with a meter as shown on page 4.

5. With wheels lifted, test each servo, then both together. Check 5 V stability, resets and board temperature at 8.4 V and 7.0 V input. Do not hold a servo stalled.

6. Verify calibrated neutral on joystick release and signal loss. Firmware cannot detect a low battery. Establish short run intervals with manual cell checks.

Before connecting USB to the S2

Unplug the battery, remove J_PWR, and disconnect both servo signal wires. Ground may remain connected. The S2 VBUS header is connected to USB power: battery disconnection alone can still let USB feed the servos or buck backward through the 5 V wire. Restore J_PWR and signals only after USB is removed.

Charge outside the car

Use a **2S LiPo balance charger** set for 4.20 V/cell (8.40 V pack). A conservative 1C setting for 300 mAh is **0.30 A**, subject to pack instructions. Connect its XT30 main lead and 3-pin balance lead as the charger requires. A 1S USB/TP4056 charger is not compatible. Charge attended on a nonflammable surface.

POC battery checks: check both cells with your multimeter before and between short, attended test runs. Do not bridge adjacent balance contacts with probe tips. End the session when either cell reaches 3.7 V at rest. Unplug when parked. Firmware has no battery sensing, alarm or low-voltage stop; periodic checks cannot guarantee protection between measurements.

Sources, image credits and build files

WEMOS: S2 Mini photo, board pinout and schematic

TI: LM2596 buck datasheet

TI: TPS63070 buck-boost datasheet

TowerPro: MG90S reference image (not proof of continuous rotation)

What is underneath the deck?

The sensing perfboard is removed. The fuse and power capacitors remain; none is a charger.

F_IN: a compact fuse candidate

Littelfuse 0251004.MXL
PICO II 251 series
4 A, very fast acting
125 V DC rated
300 A DC interrupt rating

Body: 7.11 mm long,
2.80 mm maximum diameter.
Leads and insulation add space.

251-series reference photo
Appearance / markings may vary

DigiKey: 0251004.MXL / purchase page

Littelfuse: specifications and mechanical drawing

Why these extra parts are here

Fuse: opens the battery-positive feed during a sufficiently large overcurrent, such as a wiring short. It is one-time use; replace after finding the fault. It is not low-battery protection or a precise 4 A current limiter.

No perfboard: the voltage divider, transistors, sensing resistors and ADC filter are removed. There are no GPIO3 or GPIO7 wires. Battery checks are manual; no electronic low-voltage protection remains.

Capacitors: the nearby 220 uF and 100 nF parts help smooth the 5 V supply. They do not replace an adequately sized converter.

Where the fuse connects

XT30 harness + — F_IN fuse — VBAT_SW: fused battery positive to buck IN+

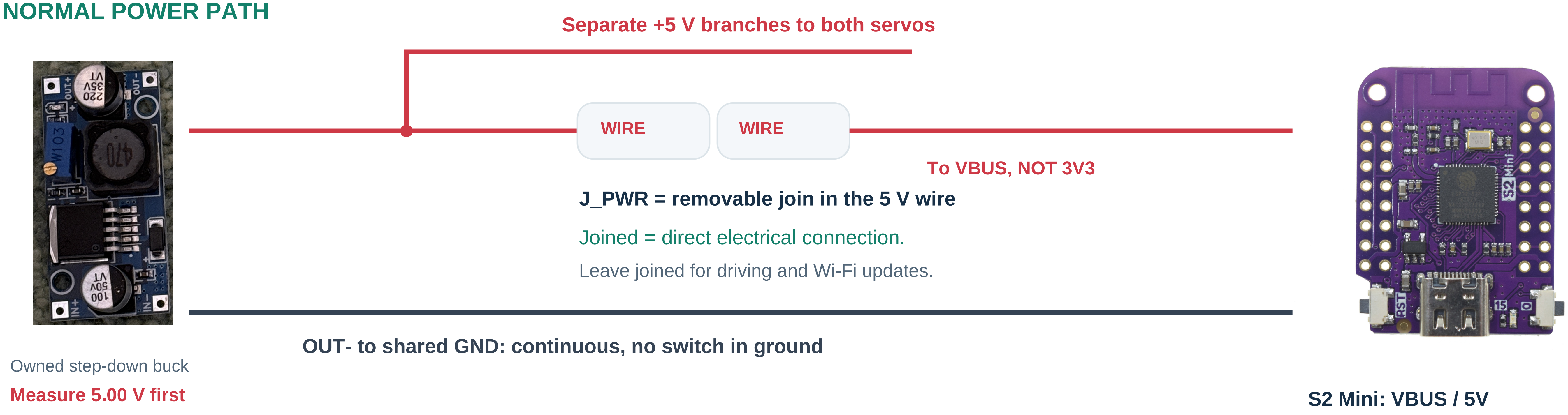
Assembly: put the fuse close to the battery connector in the positive harness lead. This axial candidate is soldered inline and insulated with strain relief; it does not require a bulky cartridge holder. Either fuse lead can face the battery. Keep motor current in the insulated power harness, away from S2 GPIOs.

Still provisional: validate 4 A against measured startup current, wire size and fault current. The updated Blender body follows the axial candidate; lead bends, insulation and loose cradle retention still need a fit check.
Photo: Littelfuse / DigiKey, reference image supplied with this product listing.

Buck 5 V goes straight to ESP32 VBUS.

J_PWR is our label for a removable join in that wire. It is not a board, regulator, or ESP32 pin.

NORMAL POWER PATH



Driving on your home Wi-Fi

Set your network in **web.private.h**. Open **http://spy-car.local/** on a computer or phone on the same network. Your computer keeps internet access. Battery, buck, J_PWR and servos stay connected; no USB cable.

Wireless update mode: current firmware

Lift the wheels. Boot normally, then hold **BOOT/0 for 3 seconds**. Driving locks off. Join **SpyCar-Update**; open **http://192.168.4.1/update**. Login: **admin**. Use your configured password for both Wi-Fi and login. Keep USB unplugged.

USB recovery and the next software step

USB initial setup / recovery: unplug XT30, separate J_PWR and unplug both servo signals first. Ground may remain connected. **Planned, not yet implemented:** updates over home Wi-Fi at spy-car.local/update, with SpyCar-Update as fallback. The wiring already supports both approaches; no extra switch is needed.